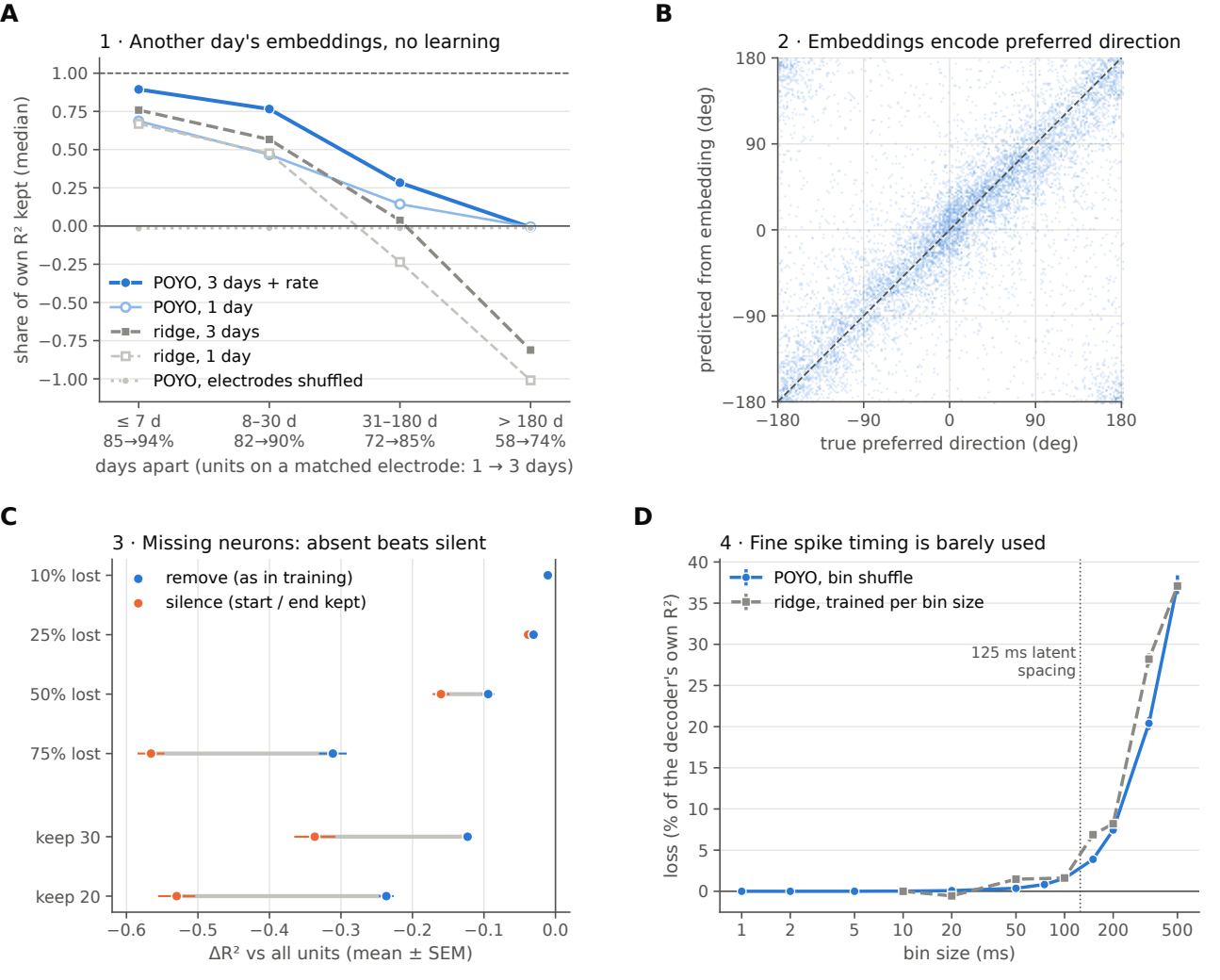
A
1 · Another day's embeddings, no learning
share of own R² kept (median)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
POYO, 3 days + rate
POYO, 1 day
ridge, 3 days
ridge, 1 day
POYO, electrodes shuffled
≤ 7 d
85→94%
8–30 d
82→90%
31–180 d
72→85%
> 180 d
58→74%
days apart (units on a matched electrode: 1 → 3 days)
B
2 · Embeddings encode preferred direction
predicted from embedding (deg)
180
90
0
−90
−180
−180
−90
0
90
180
true preferred direction (deg)
C
3 · Missing neurons: absent beats silent
remove (as in training)
silence (start / end kept)
10% lost
25% lost
50% lost
75% lost
keep 30
keep 20
−0.6
−0.5
−0.4
−0.3
−0.2
−0.1
0.0
ΔR² vs all units (mean ± SEM)
D
4 · Fine spike timing is barely used
POYO, bin shuffle
ridge, trained per bin size
125 ms latent
spacing
loss (% of the decoder's own R²)
40
35
30
25
20
15
10
5
0
1
2
5
10
20
50
100
200
500
bin size (ms)